Thus we can form an equation as $x + (x + 4) = 24$. Otherwise, $2x + 4 = 24$.

Hence an equation can be looked as algebraic expression equated to either a constant or any other algebraic expression.

Try these

Try to construct algebraic equations for the following verbal statements.

1. One third of a number plus 6 is 10
2. The sum of five times of x and 3 is 28.
3. Taking away 8 from y gives 11.
4. Perimeter of a square with side a is 16cm.
5. Venkat's mother's age is 7 years more than 3 times Venkat's age. His mother's age is 43 years.

3.6.2 Solving an equation

An equation is like a weighing balance with equal weights on both of its pans, in which case the arm of the balance are exactly horizontal.

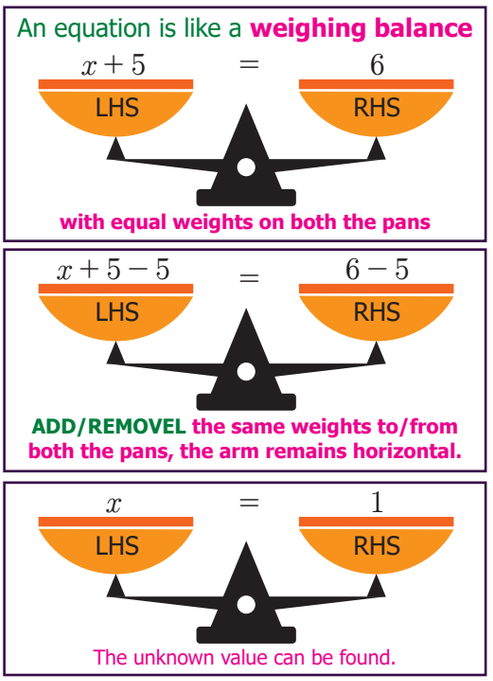


Fig.3.8

If we add the same weights to both the pans, the arms remains horizontal. In the same way, if we remove the same weights from both the pans, then also the arm remains horizontal. We use the same principle for solving an equation.

We use the following principles to separate the variables and constants thereby solving an equation.

1. If we add or subtract the same number on both sides of the equation, the value remains the same. For example, to solve $x + 5 = 12$, we have to subtract 5 on both sides, for separating the constants and variables of the equation, that is $x + 5 - 5 = 12 - 5$.

   $x + 0 = 7$ Hence $x = 7$ [since 0 is the additive identity]

Think

Why should we subtract 5 and not some other number? Why don't we add 5 on both sides? Discuss.

2. Similarly, if we multiply or divide the equation with the same number on both sides, the equation remains the same. For example, to solve the equation $5y = 20$, divide by 5 on both sides.

   Thus we have $\frac{1}{5} \times 5y = \frac{1}{5} \times 20$.

   $$y = \frac{20}{5}$$

   Therefore, $y = 4$

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